

Workshop Registration Service

Setup instructions: run it locally in about five minutes

What you need

- Node.js 20 or newer (check with: `node -v`)
- MySQL 8 running on your machine, with a user that can create databases (the examples use root)
- Two terminals, one for the backend and one for the frontend

The project has two folders, backend and frontend. Each one has its own package.json.

1. Backend (API on port 4000)

```
cd backend
npm install
cp .env.example .env
```

Open the new .env file and set DATABASE_URL to your MySQL details. Everything else can stay as it is.

```
DATABASE_URL="mysql://root:YOUR_PASSWORD@localhost:3306/workshop_registration_db"
```

If your password contains special characters, encode them. For example "@" becomes "%40". You do not need to create the database yourself, the next command does it. Then run:

```
npm run db:migrate
npm run seed
npm run dev
```

db:migrate creates the database and tables. seed adds the three locations, the Admin account and the sample data below. You should see "API listening on http://localhost:4000". To check it, open <http://localhost:4000/health> in a browser; it should show status ok.

2. Frontend (web app on port 5173)

In the second terminal:

```
cd frontend
npm install
npm run dev
```

Open <http://localhost:5173> and sign in. No .env file is needed: the frontend forwards /api calls to the backend on port 4000 by itself. If you run the backend on another port, copy frontend/.env.example to .env and change API_PROXY_TARGET.

Login details (development only)

Role	Email	Password	What to try
Admin	admin@centre.local	Admin@12345	Create staff accounts, change roles, deactivate people
Manager	manager@centre.local	Manager@12345	Add and edit workshops, register and cancel attendees
Staff (front desk)	staff@centre.local	Staff@12345	Register and cancel attendees, search workshops

These accounts exist only because the seed is set to create demo data. Change the passwords and set SEED_DEMO=false in .env before using the system for real.

Sample workshops

The seed creates five upcoming workshops, so there is something to book straight away. Dates are set relative to the day you run the seed.

Code	Title	Seats	Location	Starts
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FIT-110	Morning HIIT	3	Gampaha	Tomorrow, 7:00
POT-101	Intro to Pottery	20	Colombo	In 2 days, 10:00
CODE-201	Python for Beginners	15	Horana	In 3 days, 18:00
ART-150	Watercolour Basics	12	Colombo	In 9 days, 14:00
CODE-305	Build a Website	10	Horana	In 12 days, 18:00

A quick tour

- Sign in as Staff, open Morning HIIT (only 3 seats) and register three people. Try a fourth: you are told it is full and can add them to the waitlist.
- Cancel one of the three. The seat is freed, the waitlisted person moves up automatically, and the cancelled registration stays visible in the history.
- Sign in as Manager to add or edit a workshop. Sign in as Admin to manage accounts. Staff cannot see these options, and the backend refuses the requests even if someone tries them directly.
- Optional: with the backend running, run "npm run test:race" inside backend. It sends many bookings at once and checks that no workshop is ever overbooked.

If something goes wrong

- Cannot connect to the database: check MySQL is running and that DATABASE_URL has the right user, password and port.
- Access denied when creating the database: use a MySQL user with permission to create databases, or create an empty workshop_registration_db yourself and run the commands again.
- Port already in use: stop whatever is using 4000 or 5173, or change PORT in backend/.env.
- Want a clean start: in backend, run "npm run db:reset" (this wipes the data and seeds it again).